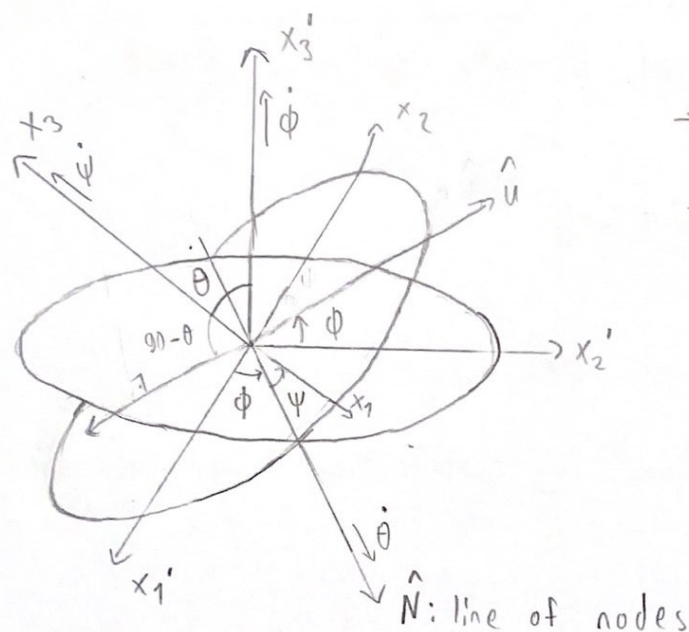


PHYS 302 - HW4

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1)



- Primed axis belong to fixed frame
- Unprimed axis belong to body frame
- Let $\hat{u}$ be the axis which is formed by rotating x_2' by ϕ degrees C.C.W
 - ↳ So $\hat{u}$ is on $x_1'-x_2'$ plane
 - ↳ perpendicular to $\hat{N}$
- $\hat{N}$ is formed by rotating x_1' ϕ degrees C.C.W
 - ↳ $\hat{N}$ is in $x_1'-x_2'$ plane

→ Notice every rotation rotates a plane and the corresponding angular velocity vector is perpendicular to that plane

↳ θ is rotation in x_3-x_3' plane so $\dot{\theta}$ is in $\hat{N}$ direction

↳ ϕ is rotation in $x_1'-x_2'$ plane so $\dot{\phi}$ is in x_3' direction

↳ ψ is rotation in x_1-x_2 plane so $\dot{\psi}$ is in x_3 direction

$$\vec{\omega}' = \omega'_1 \hat{x}_1'$$

→ It is clear that $\dot{\phi}$ won't give any contribution to ω'_1 since it's in x_3' direction which is perpendicular to x_1'

• Consider the projection of $\dot{\psi}$ onto $\hat{u}$ which is on $x_1'-x_2'$ plane

$$\text{proj}_{\hat{u}} \dot{\psi} = \dot{\psi} \cos(90-\theta) = \dot{\psi} \sin\theta$$

→ Now since $\hat{u}$ is perpendicular to $\hat{N}$ which is ϕ away from x_1' , angle between $\text{proj}_{\hat{u}} \dot{\psi}$ and x_1' is $90-\phi$

Then $\text{proj}_{x_1'} \dot{\psi} = \text{proj}_{\hat{u}} \dot{\psi} \sin \phi = \dot{\psi} \sin \theta \sin \phi$

→ And clearly projection of $\dot{\theta}$ onto x_1' is $\dot{\theta} \cos \theta$

Then $w_1' = \dot{\theta} \cos \theta + \dot{\psi} \sin \theta \sin \phi$

$\vec{w}_2' = w_2' \hat{x}_2'$

→ Again clearly $\dot{\phi}$ won't give any contribution to w_2' since it's x_3' is perpendicular to x_2'

• In order to find the proj. of $\dot{\psi}$ onto x_2' , again first project $\dot{\psi}$ onto $\hat{u}$. $\Rightarrow \text{proj}_{\hat{u}} \dot{\psi} = \dot{\psi} \sin \theta$

→ Now one can easily project $\text{proj}_{\hat{u}} \dot{\psi}$ onto x_2' by noting that angle between them is $\cos \theta$. Beware of the minus sign!!

$\text{proj}_{x_2'} \dot{\psi} = \text{proj}_{\hat{u}} \dot{\psi} \cos(180^\circ - \phi) = -\dot{\psi} \sin \theta \sin \phi$

→ And clearly projection of $\dot{\theta}$ onto x_2' is $\dot{\theta} \cos(90^\circ - \phi) = \dot{\theta} \sin \phi$

Then $w_2' = \dot{\theta} \sin \phi - \dot{\psi} \sin \theta \cos \phi$

$\vec{w}_3' = w_3' \hat{x}_3'$

→ Clearly $\dot{\theta}$ won't give any contribution to w_3' since $\hat{N}$ is perpendicular to x_3'

→ $\dot{\phi}$ is already in x_3' direction

→ $\text{proj}_{x_3'} \dot{\psi} = \dot{\psi} \cos \theta$

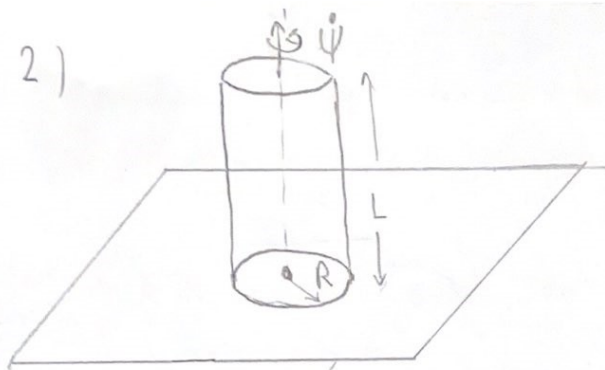
Then $w_3' = \dot{\phi} + \dot{\psi} \cos \theta$

$w_1' = \dot{\theta} \cos \phi + \dot{\psi} \sin \theta \sin \phi$

$w_2' = \dot{\theta} \sin \phi - \dot{\psi} \sin \theta \cos \phi$

$w_3' = \dot{\phi} + \dot{\psi} \cos \theta$

2)



* Choose a body-fixed axis xyz such that axes are eigenvectors of the moment of inertia tensor and $\hat{z}$ is along the symmetry axis of the cylinder through its length

→ Since we want the cylinder to stay upright, we will investigate the case in which symmetric top is not precessing, and it only has a spin $\dot{\psi} \Rightarrow \vec{\omega} = \dot{\psi} \hat{z}$.

→ From our previous studies where we have considered the stability of the symmetric top around $\theta=0$:

$$\tilde{V}_{\text{eff}}(\theta) = \frac{(a-b \cos \theta)^2}{\sin^2 \theta} + c \cos \theta \quad \text{where}$$

$$a = \frac{P_{\phi}}{\sqrt{2} I}$$

$$b = \frac{P_{\psi}}{\sqrt{2} I}$$

$$c = \frac{Mgh}{I}$$

↳ By expanding $\tilde{V}_{\text{eff}}(\theta)$ around $\theta=0$ and requiring $\tilde{V}_{\text{eff}}(\theta)$ to stay small under perturbation. $\left(\frac{d^2 \tilde{V}_{\text{eff}}(\theta)}{d\theta^2} > 0 \right)$, one can

show that we need $a=b$ and $b^2 > 2c$ for stability.

$$\tilde{V}_{\text{eff}}(\theta) \Big|_{\theta=0} \approx \frac{(a-b)^2}{\theta^2} + \frac{1}{3} [(a-b)(a+2b) + c] + \frac{1}{60} A \theta^2 + \frac{B}{1512} \theta^4 + \dots$$

↳ $a=b$ required for $f(\theta)$ not to explode to infinity

$$\tilde{V}_{\text{eff}}(\theta) \Big|_{a=b} = c + \frac{1}{4} (b^2 - 2c) \theta^2 + \frac{1}{24} \underbrace{(b^2 + c)}_{>0} \theta^4 + \dots$$

$b^2 > 2c$ required for $\frac{d^2 \tilde{V}_{\text{eff}}(\theta)}{d\theta^2} > 0$

We can achieve $a=b$ by setting $\dot{\phi}(0)=0$ which is automatically satisfied for our analysis as $\theta=0$

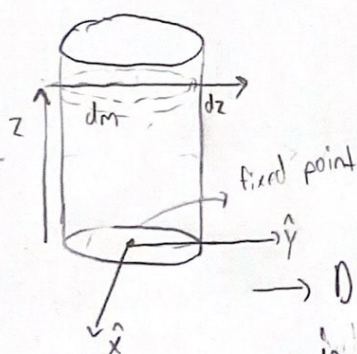
$$\rightarrow b^2 > 2c \quad \text{corresponds to} \quad \frac{P_\psi^2}{2I^2} > 2 \frac{Mgh}{I}$$

$$P_\psi = I_3(\dot{\psi} + \dot{\phi} \cos \theta) = I_3 \dot{\psi} \quad \text{as } \dot{\phi}(t)=0 \quad (\text{no precession})$$

↳ Here I_3 is the moment of inertia along symmetry axis of the cylinder ($\hat{z}$ direction) w.r.t. the fixed point

$$I_3^2 \dot{\psi}^2 > 4IMgh \Rightarrow \dot{\psi}^2 > \frac{4MghI}{I_3^2} \rightarrow \text{moment of inertia along x or y axes.}$$

→ So we need its spin to be at least $\frac{4MghI}{I_3^2}$. One can achieve this by giving an initial spin $\dot{\psi}(0) > \frac{4MghI}{I_3^2}$ if there is no friction.



$$I = \int \left(\underbrace{\frac{1}{4} dm R^2}_{\text{moment of inertia of a disc along its diameter}} + \underbrace{dm z^2}_{M \text{ term}} \right) \Rightarrow \text{from parallel axis theorem}$$

→ Divide the cylinder into discs of thickness dz , mass dm in which z is essentially constant throughout this cylinder.

$$dm = \frac{M}{\pi R^2 L} \pi R^2 dz = \frac{M}{L} dz$$

$$I = \frac{M}{L} \int_0^L \left(\frac{1}{4} R^2 + z^2 \right) dz = \frac{M}{L} \left[\frac{R^2}{4} z + \frac{z^3}{3} \right]_0^L = \frac{MR^2}{4} + \frac{ML^2}{3}$$

→ We've calculated I and I_3 can easily shown to be:

$$I_3 = \frac{1}{2} MR^2 \Rightarrow \text{Note that } I_3 \text{ w.r.t. CM is same as } I_3 \text{ w.r.t. fixed point since they're both on the same axis.}$$

→ Time to put all quantities into their place:

• Notice $h = \frac{L}{2}$, $R = \frac{D}{2} \rightarrow \text{diameter}$

$$I_3 = \frac{MD^2}{8}, \quad I = \frac{MD^2}{16} + \frac{ML^2}{3}$$

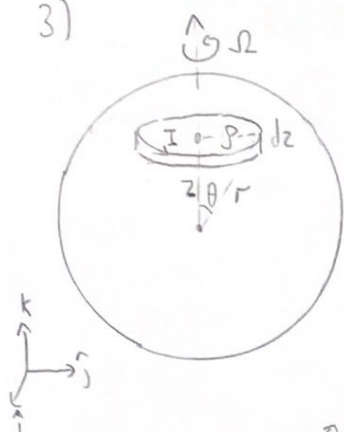
$$\dot{\psi}^2 > \frac{4Mg \frac{L}{2} M (D^2/16 + L^2/3)}{\frac{M^2 D^4}{64}} = \frac{8gL}{D^2} \left(1 + \frac{8L^2}{3D^2} \right)$$

$$\dot{\psi}^2 > \frac{8 \times 9.89 \text{ m/s}^2 \times 0.15 \text{ m}}{10^{-4} \text{ m}^2} \left(1 + \frac{8 \times 15^2}{3} \right) = 71\,326\,680 \left(\frac{\text{rad}}{\text{s}} \right)^2$$

$$\dot{\psi} > 8445.51 \text{ rad/s}$$

↳ Minimum spin for pen to stay upright is actually quite fast!

3)



Consider a uniformly charged sphere of radius \$R\$ and charge \$Q\$ rotating with \$\Omega\$.

In this sphere consider a current loop (essentially a thin strip) of radius \$\rho\$, height \$dz\$ and width \$dp\$. Since thickness and height is infinitesimally small, we can model it to be an Amperian loop with current:

$$I = \frac{\Delta q}{\Delta t} = \frac{\overset{\text{volume}}{dV} \overset{\text{charge density}}{\rho_{\text{charge}}}}{T \rightarrow \text{period}}$$

$$dV = 2\pi\rho \, dz \, dp, \quad \rho_{\text{charge}} = \frac{Q}{\frac{4}{3}\pi R^3}, \quad T = \frac{2\pi}{\Omega}$$

* We know from electromagnetic theory that magnetic dipole moment of a closed loop is:

$$\vec{\mu} = I \int d\vec{A} \Rightarrow d\vec{\mu} = I \pi \rho^2 \hat{k}$$

$$d\vec{\mu} = \frac{\Omega}{2\pi} \frac{Q}{\frac{4}{3}\pi R^3} \pi \rho^2 2\pi \rho \, dz \, dp \, \hat{k} = \frac{3Q\Omega}{4R^3} \rho^3 \, dp \, dz \, \hat{k}$$

$$\vec{\mu}_T = \int d\vec{\mu} = \frac{3Q\Omega \hat{k}}{4R^3} \int_{-R}^R \int_0^{R\sqrt{R^2-z^2}} \rho^3 \, dp \, dz = \frac{3Q\Omega \hat{k}}{4R^3} \int_{-R}^R \frac{(R^2-z^2)^2}{4} \, dz$$

$$\begin{aligned} \vec{\mu}_T &= \frac{3Q\Omega \hat{k}}{16R^3} \int_{-R}^R (R^4 - 2R^2 z^2 + z^4) \, dz = \frac{3Q\Omega \hat{k}}{16R^3} \left[R^4 z - \frac{2R^2}{3} z^3 + \frac{z^5}{5} \right]_{-R}^R \\ &= \frac{3Q\Omega \hat{k}}{16R^3} \left(2R^5 - \frac{4R^5}{3} + \frac{2R^5}{5} \right) = \boxed{\frac{1}{5} Q \Omega R^2 \hat{k}} \end{aligned}$$

Now we know that moment of inertia of a rotating homogeneous rigid sphere about any of its symmetry axis is $I = \frac{2}{5} MR^2$

Then in our case total angular momentum of the sphere is simply:

$$\vec{L}_T = \frac{2}{5} MR^2 \Omega \hat{k},$$

→ If one were to pay attention to the ratio of the magnetic dipole moment to the angular momentum:

$$\frac{\vec{\mu}_T}{\vec{L}_T} = \frac{\frac{1}{3} QR^2 \Omega \hat{k}}{\frac{2}{5} MR^2 \Omega \hat{k}} = \frac{Q}{2M} =$$

→ This ratio is known as the "gyromagnetic ratio" and it is the same for rotating bodies (uniformly charged & homogeneous mass distribution) regardless of the geometry.

• For electrons: $\frac{\vec{\mu}_e}{\vec{L}_e} = \frac{e}{2m} = \gamma$ is still pertinent
spin of electrons

Now we know that $\vec{\tau} = \vec{\mu} \times \vec{B}$. Then

$$\vec{\tau} = \frac{d\vec{L}}{dt} = \vec{\mu} \times \vec{B} = \left(\frac{Q}{2m} \vec{L} \right) \times \vec{B} = \vec{L} \times \left(\frac{Q}{2m} \vec{B} \right)$$

$$\frac{d\vec{L}}{dt} = \left(-\frac{Q}{2m} \vec{B} \right) \times \vec{L} = \vec{\omega}_p \times \vec{L} \quad \square$$

where $\vec{\omega}_p = -\frac{Q}{2m} \vec{B}$

4) A symmetric top with $I_1 = I_2 = I \neq I_3$ with one point fixed in a gravitational field.

$$\theta(0) = \pi/3, \quad \dot{\theta}(0) = 0, \quad \dot{\phi}(0) = 2\sqrt{\frac{Mgh}{3I}}, \quad \dot{\psi}(0) = \left(\frac{3I - I_3}{I_3}\right) \sqrt{\frac{Mgh}{3I}}$$

$$P_\phi = (I \sin^2 \theta_0 + I_3 \cos^2 \theta_0) \dot{\phi}(0) + I_3 \cos \theta_0 \dot{\psi}(0)$$

$$= \left(\frac{3}{4}I + \frac{1}{4}I_3\right) 2\sqrt{\frac{Mgh}{3I}} + \frac{1}{2}(3I - I_3) \sqrt{\frac{Mgh}{3I}} = \sqrt{\frac{Mgh}{3I}} \left(\frac{3}{2}I + \frac{I_3}{2} + \frac{3I}{2} - \frac{I_3}{2}\right)$$

$$P_\phi = 3I \sqrt{\frac{Mgh}{3I}}$$

$$P_\psi = I_3 (\dot{\psi}(0) + \dot{\phi}(0) \cos \theta_0) = \sqrt{\frac{Mgh}{3I}} (3I - I_3 + I_3) \Rightarrow P_\psi = 3I \sqrt{\frac{Mgh}{3I}}$$

$$V_{\text{eff}}(\theta) = \frac{(P_\phi - P_\psi \cos \theta)^2}{2I \sin^2 \theta} + Mgh \cos \theta = \frac{9I^2 \frac{Mgh}{3I} (1 - \cos \theta)^2}{2I \sin^2 \theta} + Mgh \cos \theta$$

$$V_{\text{eff}}(\theta) = Mgh \left[\cos \theta + \frac{3(1 - \cos \theta)^2}{2 \sin^2 \theta} \right]$$

* From our previous studies we've found that, the quantity

$$E' = \frac{1}{2} I \dot{\theta}^2 + \underbrace{\frac{(P_\phi - P_\psi \cos \theta)^2}{2I \sin^2 \theta} + Mgh \cos \theta}_{V_{\text{eff}}(\theta)} \quad \text{was conserved}$$

At $t=0$, $\dot{\theta}(0)=0$. So one can find E' to be:

$$E' = 0 + V_{\text{eff}}(\theta_0) = Mgh \left[\frac{1}{2} + \frac{3}{2} \frac{1/4}{3/4} \right] = Mgh //$$

Let $u = \cos\theta$ and write $V_{\text{eff}}(\theta)$ in terms of u :

$$V_{\text{eff}}(u) = Mgh \left[u + \frac{3(1-u)^2}{2(1-u^2)} \right]$$

→ Notice $\dot{u} = -\sin\theta \dot{\theta} \Rightarrow \dot{\theta}^2 = \frac{\dot{u}^2}{\sin^2\theta} = \frac{\dot{u}^2}{1-u^2}$

$$E = Mgh = \frac{1}{2} I \frac{\dot{u}^2}{1-u^2} + Mgh \left[u + \frac{3(1-u)^2}{2(1-u^2)} \right]$$

$$\frac{1}{2} I \frac{\dot{u}^2}{1-u^2} = Mgh \left[1-u - \frac{3}{2} \frac{(1-u)^2}{(1-u^2)} \right] = Mgh(1-u) \left[1 - \frac{3}{2} \left(\frac{1-u}{1+u} \right) \right]$$

$$\dot{u}^2 = \frac{Mgh}{I} (1-u) \left[2(1+u) - 3(1-u) \right] = \frac{Mgh}{I} (1-u) \underbrace{(-1 + 3u - 2u^2)}_{(1-u)(2u-1)}$$

$$\boxed{\dot{u}^2 = \frac{Mgh}{I} (1-u)^2 (2u-1)}$$

Now integrals involved in finding the solution is a bit complicated but nothing we can't handle:

First let $V = \frac{1}{u} = \sec\theta$, $\frac{dV}{dt} = \dot{V} = -\frac{1}{u^2} \frac{du}{dt} \Rightarrow \dot{u} = -u^2 \dot{V} = -\frac{\dot{V}}{V^2}$

Rewriting the equation in terms of V :

$$\frac{1}{V^4} \dot{V}^2 = \frac{Mgh}{I} \left(1 - \frac{1}{V} \right)^2 \left(\frac{2}{V} - 1 \right) = \frac{Mgh}{I} \frac{1}{V^4} (V-1)^2 (2V-V^2)$$

$$\frac{\dot{V}^2}{(V-1)^2 (2V-V^2)} = \frac{Mgh}{I}, \quad \frac{\dot{V}}{(V-1) \sqrt{(2V-V^2)^2}} = \pm \sqrt{\frac{Mgh}{I}}$$

we'll decide the sign later.

$$\frac{dV}{(V-1) \sqrt{1-(V-1)^2}} = \pm \sqrt{\frac{Mgh}{I}} dt \quad \text{by noting } 1-(V-1)^2 = 2V-V^2$$

Letting $V-1 = \gamma$
 $dV = d\gamma$

$$\frac{d\gamma}{\gamma \sqrt{1-\gamma^2}} = \pm \sqrt{\frac{Mgh}{I}} dt$$

Letting $\gamma = \operatorname{sech}(\beta)$ and using the identity $\operatorname{sech}^2 \beta + \tanh^2 \beta = 1$
 $d\gamma = -\operatorname{sech}(\beta) \tanh(\beta) d\beta$

$$\int \frac{-\operatorname{sech}(\beta) \tanh(\beta) d\beta}{\operatorname{sech}(\beta) \underbrace{\sqrt{1-\operatorname{sech}^2(\beta)}}_{\tanh(\beta)}} = \int -d\beta = \pm \sqrt{\frac{Mgh}{I}} dt$$

→ Integrating, we get $\beta = \mp \sqrt{\frac{Mgh}{I}} t + C$

$$\gamma = \operatorname{sech}\left(\mp \sqrt{\frac{Mgh}{I}} t + C\right) = \operatorname{sech}\left(\sqrt{\frac{Mgh}{I}} t + C\right) \quad \text{since } \operatorname{sech}(x) \text{ is an even function}$$

$$V = \gamma + 1 = \operatorname{sech}\left(\sqrt{\frac{Mgh}{I}} t + C\right) + 1 = \sec \theta$$

At $t=0 \Rightarrow \sec \theta_0 = 2 = 1 + \operatorname{sech}(C) \Rightarrow \operatorname{sech}(C) = 1$

* $C=0$ since $\operatorname{sech}(x)$ is 1 only at $x=0$

$$\boxed{\sec \theta = 1 + \operatorname{sech}\left(\sqrt{\frac{Mgh}{I}} t\right)}$$

$$5) \quad \dot{\phi} = \frac{P_{\phi} - P_{\psi} \cos \theta}{I \sin^2 \theta} = 3 \sqrt{\frac{Mgh}{3I}} \left(\frac{1 - \cos \theta}{1 - \cos^2 \theta} \right) = 3 \sqrt{\frac{Mgh}{3I}} \left(\frac{\sec^2 \theta - \sec \theta}{\sec^2 \theta - 1} \right)$$

$$\dot{\phi} = 3 \sqrt{\frac{Mgh}{3I}} \left[\frac{1 + 2 \operatorname{sech}(\cdot) + \operatorname{sech}^2(\cdot) - 1 - \operatorname{sech}(\cdot)}{1 + 2 \operatorname{sech}(\cdot) + \operatorname{sech}^2(\cdot) - 1} \right]$$

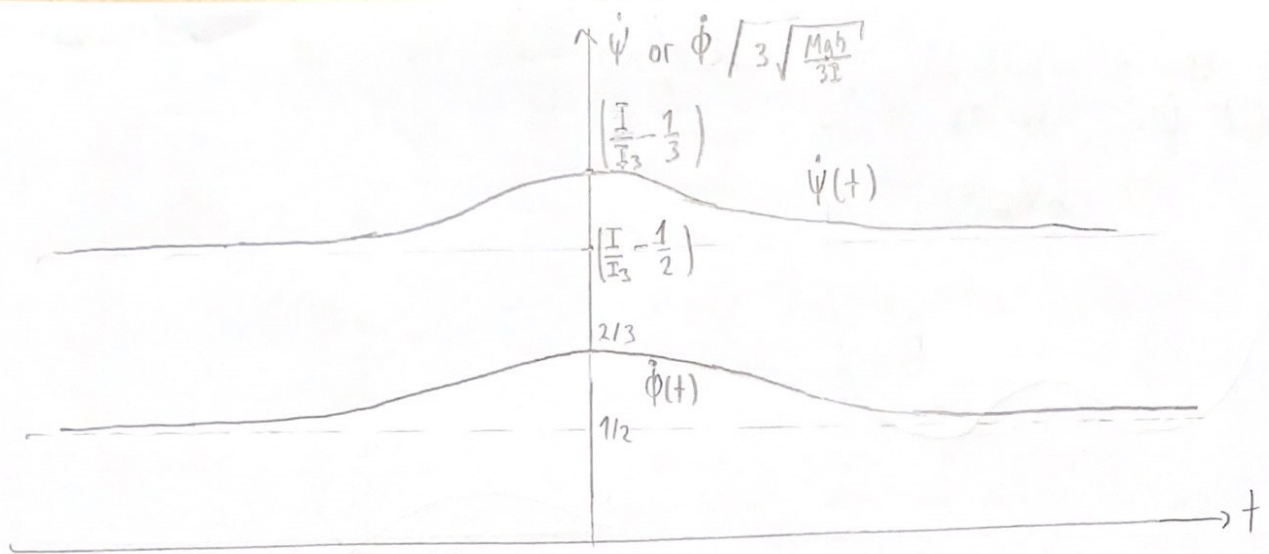
$$= 3 \sqrt{\frac{Mgh}{3I}} \left[\frac{1 + \operatorname{sech}(\cdot)}{2 + \operatorname{sech}(\cdot)} \right] \Rightarrow$$

$$\dot{\phi}(t) = 3 \sqrt{\frac{Mgh}{3I}} \left[\frac{1 + \operatorname{sech}\left(\sqrt{\frac{Mgh}{I}} t\right)}{2 + \operatorname{sech}\left(\sqrt{\frac{Mgh}{I}} t\right)} \right]$$

$$\dot{\psi} = \frac{P_{\psi}}{I_3} - \frac{(P_{\phi} - P_{\psi} \cos \theta) \cos \theta}{I \sin^2 \theta} = 3 \sqrt{\frac{Mgh}{3I}} \left(\frac{I}{I_3} + \frac{\cos^2 \theta - \cos \theta}{1 - \cos^2 \theta} \right)$$

$$\dot{\psi} = 3 \sqrt{\frac{Mgh}{3I}} \left(\frac{I}{I_3} + \frac{1 - \sec \theta}{\sec^2 \theta - 1} \right) = 3 \sqrt{\frac{Mgh}{3I}} \left[\frac{I}{I_3} + \frac{1 - 1 - \operatorname{sech}(\cdot)}{1 + 2 \operatorname{sech}(\cdot) + \operatorname{sech}^2(\cdot) - 1} \right]$$

$$\dot{\psi}(t) = 3 \sqrt{\frac{Mgh}{3I}} \left[\frac{I}{I_3} - \frac{1}{2 + \operatorname{sech}\left(\sqrt{\frac{Mgh}{I}} t\right)} \right]$$

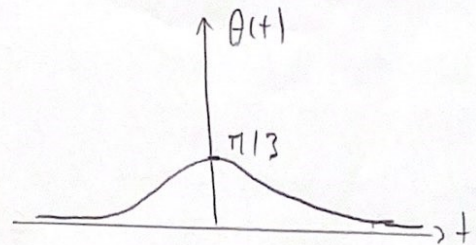


→ We see that both $\dot{\psi}$ and $\dot{\phi}$ decrease to their steady values after a short while.

$$\lim_{t \rightarrow \infty} \dot{\phi}(t) = 3\sqrt{\frac{Mgh}{3I}} \lim_{t \rightarrow \infty} \frac{1 + \text{sech}(t)}{2 + \text{sech}(t)} = \frac{3}{2} \sqrt{\frac{Mgh}{3I}}$$

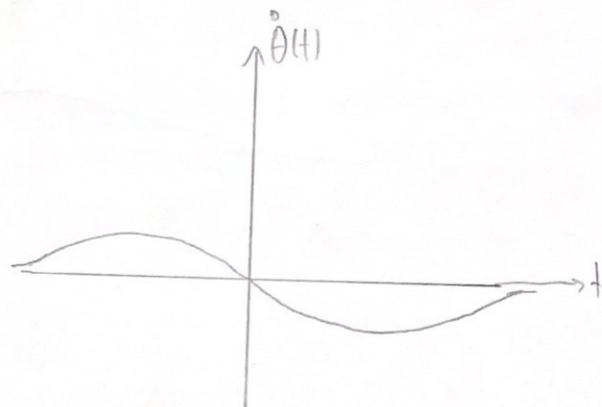
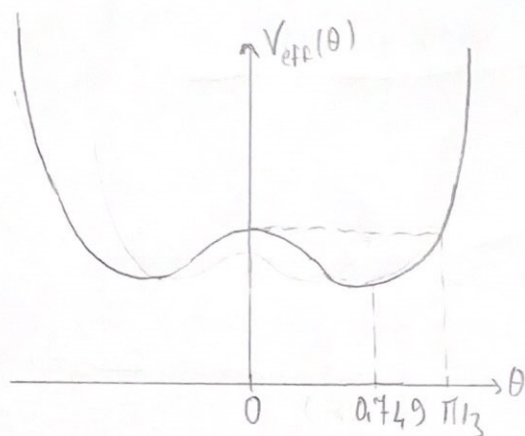
$$\lim_{t \rightarrow \infty} \dot{\psi}(t) = 3\sqrt{\frac{Mgh}{3I}} \lim_{t \rightarrow \infty} \left[\frac{I}{I_3} - \frac{1}{2 + \text{sech}\left(\sqrt{\frac{Mgh}{I}}t\right)} \right] = 3\sqrt{\frac{Mgh}{3I}} \left(\frac{I}{I_3} - \frac{1}{2} \right)$$

$$\lim_{t \rightarrow \infty} \theta(t) = \arccos\left(\frac{1}{1 + \text{sech}x}\right) = 0$$



$$\lim_{t \rightarrow \infty} \theta(t) = 0$$

To better understand the behaviour of the top plot $V_{\text{eff}}(\theta)$ and $\dot{\theta}(t)$ curves

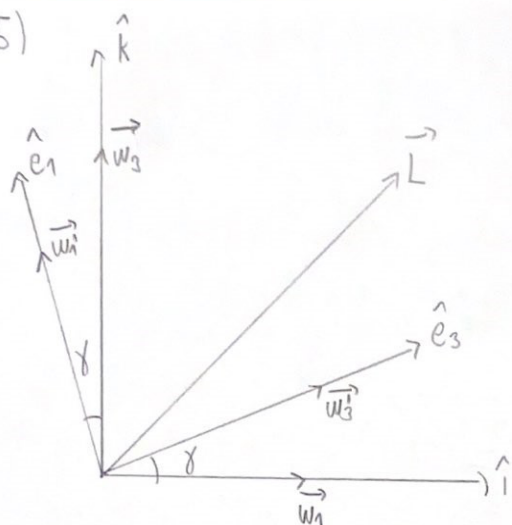


* Clearly seen from the graph of $V(\theta)$, where we leave the top $\theta = \pi/3$ is not a stable position. Top's axis will move upwards trying to reach the minimum potential point. But when it reaches this point, it now has a nonzero $\dot{\theta}(t)$ which will drive the top towards $\theta = 0$. The key point is $V_{\text{eff}}(\pi/3) = V_{\text{eff}}(0)$, so as $t \rightarrow \infty$ top's $\dot{\theta}$ must go to zero and the top should reach the unstable point $\theta = 0$. Any slight perturbation is enough to repel the top from $\theta = 0$ position. If the top were to be left of from an angle $\theta' > \pi/3$ it would basically oscillate between $-\theta'$ and $+\theta'$.

Note that $\beta^2 < 2\gamma$, so it is expected that $\theta = 0$ is not a stable point although $\alpha = \beta$.

→ In this very special setup though since there is absolutely no friction and $V(\pi/3) = V(0)$, θ is expected to get ever so closer to $\theta = 0$ as $t \rightarrow \infty$ and $\dot{\theta} \rightarrow 0$ there

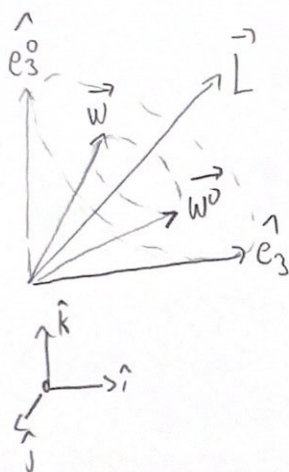
5)



* Let $\vec{w}_3$ and $\vec{w}_1$ be the angular velocity of the coin right after it was tossed into the air

* Let $\hat{e}_3$ be the body axis at an instant where it deviates the most from $\hat{k}$ axis

* We have chosen the direction of initial angular velocity about the diameter to be $\hat{i}$



* We know from our previous study that in the absence of any torque, $\vec{L}$ will stay constant. Body axis and angular velocity vectors will rotate around $\vec{L}$. At any instant $\vec{L}$, $\vec{w}$, $\hat{e}_3$ are coplanar and relationship between them is given by:

$$\frac{\vec{L}}{I} = \vec{w} + \Omega \hat{e}_3$$

→ Then $\vec{w}_3$ is the angular velocity along the body axis of the coin when it deviates the most from $\hat{k}$ and $\vec{w}_1$ is the angular velocity about the coin's diameter at that time. That's why $\vec{w}_1$, $\vec{w}_3$ are still in the $\hat{i}$ - $\hat{k}$ plane.

Initially we have:

$$\vec{L} = I w_1 \hat{i} + I_3 w_3 \hat{k}$$

$$T = \frac{1}{2} (I w_1^2 + I_3 w_3^2)$$

where I is moment of inertia about the coin's diameter, I_3 " " symmetry axis

At the instant where $\hat{e}_3$ deviates the most from $\hat{k}$

$$\vec{L} = I \omega_1' \hat{e}_1 + I_3 \omega_3' \hat{e}_3$$

$$T = \frac{1}{2} (I \omega_1'^2 + I_3 \omega_3'^2)$$

→ Since there is no force doing work on the rotational degree of freedom (gravity only affects the motion of CM not the rotation of the rigid body) and no net torque acting on the rigid body, angular momentum and rotational kinetic energy will be conserved.

$$\vec{L} \cdot \hat{e}_1 = I \omega_1' = I \omega_1 \underbrace{\hat{i} \cdot \hat{e}_1}_{-\sin \gamma} + I_3 \omega_3 \underbrace{\hat{k} \cdot \hat{e}_1}_{\cos \gamma}$$

$$\vec{L} \cdot \hat{e}_3 = I_3 \omega_3' = I \omega_1 \underbrace{\hat{i} \cdot \hat{e}_3}_{\cos \gamma} + I_3 \omega_3 \underbrace{\hat{k} \cdot \hat{e}_3}_{\sin \gamma}$$

$$\omega_1' = -\omega_1 \sin \gamma + \frac{I_3}{I} \omega_3 \cos \gamma, \quad \omega_3' = \frac{I}{I_3} \omega_1 \cos \gamma + \omega_3 \sin \gamma$$

$$I \omega_1'^2 + I_3 \omega_3'^2 = \frac{I_3^2}{I} \omega_3^2 \cos^2 \gamma + I \omega_1^2 \sin^2 \gamma - 2 I_3 \omega_1 \omega_3 \sin \gamma \cos \gamma$$

$$+ \frac{I^2}{I_3} \omega_1^2 \cos^2 \gamma + I_3 \omega_3^2 \sin^2 \gamma + 2 I \omega_1 \omega_3 \sin \gamma \cos \gamma$$

$$= I \omega_1^2 + I_3 \omega_3^2$$

$$2 \omega_1 \omega_3 \sin \gamma \cos \gamma (I - I_3) + \frac{I_3^2}{I} \omega_3^2 \cos^2 \gamma + \frac{I^2}{I_3} \omega_1^2 \cos^2 \gamma = I \omega_1^2 \cos^2 \gamma + I_3 \omega_3^2 \cos^2 \gamma$$

$$2 \omega_1 \omega_3 \sin \gamma (I - I_3) = \cos \gamma \left[I \omega_1^2 \left(1 - \frac{I}{I_3} \right) + I_3 \omega_3^2 \left(1 - \frac{I_3}{I} \right) \right]$$

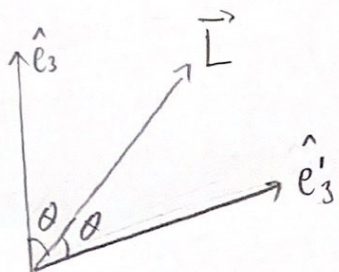
$$= \cos \gamma (I - I_3) \left(\frac{I_3}{I} \omega_3^2 - \frac{I}{I_3} \omega_1^2 \right)$$

$$\tan \gamma = \frac{1}{2\omega_1 \omega_3} \left(\frac{I_3}{I} \omega_3^2 - \frac{I}{I_3} \omega_1^2 \right)$$

→ Now in order for an observer looking from above to see only 1 face of the coin $\gamma \geq 0$. Then $\tan \gamma \geq 0$

$$\frac{I_3}{I} \omega_3^2 - \frac{I}{I_3} \omega_1^2 \geq 0 \quad \Rightarrow \quad \boxed{\frac{\omega_3}{\omega_1} \geq \frac{I}{I_3}}$$

Alternative approach



→ We have concluded in our previous study that $\hat{e}_3$ and $\vec{w}$ should rotate around the constant $\vec{L}$ with $\Omega = \left(\frac{I_3 - I}{I} \right) \omega_3^0$. So the angle between $\vec{L}$ and $\hat{e}_3$ should not change

→ Then similar to our previous argument:

$2\theta \leq 90^\circ$, $\theta \leq 45^\circ$ to only see 1 face of the coin.

$$\left. \begin{array}{l} L \cos \theta = I_3 \omega_3 \\ L \sin \theta = I \omega_1 \end{array} \right\} \tan \theta = \frac{I \omega_1}{I_3 \omega_3} \leq 1$$

↳ projections of L onto the body axis and diameter axis.

$$\frac{\omega_3}{\omega_1} \geq \frac{I}{I_3} \quad \text{as before} \quad ,$$

6)

* For the general rigid body motion :

$$(\vec{L})_a = \sum_{b=1}^3 I_{ab} \omega_b, \quad T = \frac{1}{2} \vec{L} \cdot \vec{\omega}$$

→ Let's choose, for the sake of simplicity, a reference frame rotating with $\vec{\omega}$. Since we know that $I' = O I O^T$ we can choose the eigenvectors of the moment of inertia tensor as the basis for our rotating reference frame. Then moments of inertia becomes the eigenvalues of the moment of inertia tensor.

$$I = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix}$$

→ Then in principal axis component angular momentum and kinetic energy will read as :

$$\vec{L} = I_1 \omega_1 \hat{x}_1' + I_2 \omega_2 \hat{x}_2' + I_3 \omega_3 \hat{x}_3' \quad (\text{where } \hat{x}_i' \text{'s are principal axes})$$

$$T = \frac{1}{2} I_1 \omega_1^2 + \frac{1}{2} I_2 \omega_2^2 + \frac{1}{2} I_3 \omega_3^2$$

→ If this rigid body is torque free and for free, total energy will correspond to kinetic energy $E = T$. Furthermore E and $|\vec{L}|^2$ is conserved.

$$\left. \begin{aligned} 2E &= I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2 : \text{constant} \\ |\vec{L}|^2 &= I_1^2 \omega_1^2 + I_2^2 \omega_2^2 + I_3^2 \omega_3^2 : \text{constant} \end{aligned} \right\} \text{with } I_3 > I_2 > I_1 > 0$$

$$\left\{ \begin{aligned} 2EI_1 - |\vec{L}|^2 &= (I_1 - I_2) I_2 \omega_2^2 + (I_1 - I_3) I_3 \omega_3^2 & (1) \\ 2EI_2 - |\vec{L}|^2 &= (I_2 - I_1) I_1 \omega_1^2 + (I_2 - I_3) I_3 \omega_3^2 & (2) \\ 2EI_3 - |\vec{L}|^2 &= (I_3 - I_1) I_1 \omega_1^2 + (I_3 - I_2) I_2 \omega_2^2 & (3) \end{aligned} \right.$$

Mathematical Preliminary

Consider the equation where A, B are of the same sign:

$$A \alpha^2(t) + B \beta^2(t) = C \text{ (some constant)}$$

→ Notice that $\alpha(t)$ and $\beta(t)$ are bounded since if (per say) $\alpha(t)$ grows without a bound $\beta(t)$ can't compensate for the growth of α to make sure C is constant.

A possible (although not only) solution for $\alpha(t)$ and $\beta(t)$ are

$$\alpha(t) = \sqrt{\frac{C}{A}} \cos(t + \phi) \quad , \quad \beta(t) = \sqrt{\frac{C}{B}} \sin(t + \phi)$$

which are bounded.

Now - consider the equation: where A and B are of the same sign

$$A \alpha^2(t) - B \beta^2(t) = C \text{ (const)}$$

→ Contrary to the last situation, $\alpha(t)$ and $\beta(t)$ (in general) are unbounded. If $\alpha^2(t)$ grows unboundedly, $\beta(t)$ must grow as well to make sure C stays constant

A possible (although not only) solution for $\alpha(t)$ and $\beta(t)$ are:

$$\alpha(t) = \sqrt{\frac{C}{A}} \cosh(t + \phi) \quad \beta(t) = \sqrt{\frac{C}{B}} \sinh(t + \phi)$$

which grows unboundedly with t .

Case 1: $w_1^0 \neq 0$, $w_2^0 = w_3^0 = 0$

Let us pick $w_2^0 \sim w_3^0$ order ϵ a small number.

Equation (1) reads as: $\underbrace{(I_1 - I_2)I_2 w_2^2}_{<0} + \underbrace{(I_1 - I_3)I_3 w_3^2}_{<0} = 2EI_1 - |\vec{L}|^2$; constant

↳ We've concluded in mathematical preliminary that these type of systems are bounded since coefficients of w_2^2 and w_3^2 are of the same sign.

→ So w_2 and w_3 will stay at the order of $\epsilon \Rightarrow$ bounded

From either (2) or (3):

$$2EI_2 - |\vec{L}|^2 = (I_2 - I_1)I_1 w_1^2 + (I_2 - I_3)I_3 w_3^2$$

Since w_3 is of the order ϵ , $w_3^2 \approx 0$. Then w_1 is constant

$$\dot{w}_1 \approx 0, \quad w_1(t) \approx w_1^0$$

Case 2: $w_3^0 \neq 0$, $w_1^0 = w_2^0 = 0$

Very similar to the case above:

Equation (3) reads as $\underbrace{(I_3 - I_1)I_1 w_1^2}_{>0} + \underbrace{(I_3 - I_2)I_2 w_2^2}_{>0} = 2EI_3 - |\vec{L}|^2$; const

↳ System is bounded since coefficients for w_1^2 and w_2^2 are of the same sign. w_2 and w_3 will stay at the order of $\epsilon \Rightarrow$ bounded

From either (1) or (2):

$$2EI_1 - |\vec{L}|^2 = (I_1 - I_2)I_2 w_2^2 + (I_1 - I_3)I_3 w_3^2$$

w_2 is of the order ϵ , $w_2^2 \approx 0$. Then w_3 is constant

$$\dot{w}_3 \approx 0, \quad w_3(t) \approx w_3^0$$

Case 3: $w_2^0 \neq 0$, $w_1^0 = w_3^0 = 0$

Let us again pick $w_1^0 \sim w_3^0$ order ϵ .

Eq. (2) reads as: $\underbrace{(I_2 - I_1)I_1 w_1^2}_{>0} + \underbrace{(I_2 - I_3)I_3 w_3^2}_{<0} = 2E I_2 - |\vec{L}|^2 : \text{const}$

↓ We've concluded in mathematical preliminary that these types of systems are unbounded since coefficients of w_1^2 and w_3^2 are of opposite sign

→ So w_1 and w_3 will grow unboundedly

From either (1) or (3):

$$2E I_3 - |\vec{L}|^2 = (I_3 - I_1)I_1 w_1^2 + (I_3 - I_2)I_2 w_2^2$$

↳ since w_1 grows unboundedly and doesn't stay at the order of ϵ

$$w_2 \neq 0, \quad w_2(t) \neq w_2^0$$

Conclusion

Rotations starting purely (or approximately) about $\hat{e}_1$ or $\hat{e}_3$ (the axis of smallest or largest moment of inertia) are stable

However rotations starting from $\hat{e}_2$ (the axis that has middle moment of inertia) are unstable

→ This statements constitutes the tennis racket theorem.

Alternative Approach:

Given that $I_3 > I_2 > I_1 > 0$, constraints of w_1, w_2, w_3 :

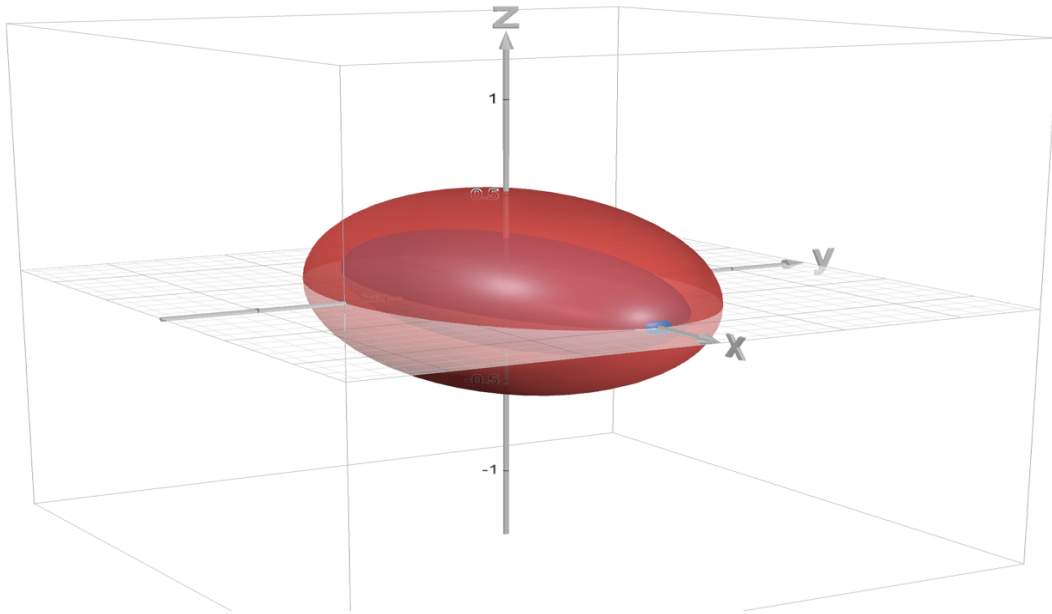
$$|\vec{L}|^2 = I_1^2 w_1^2 + I_2^2 w_2^2 + I_3^2 w_3^2$$

$$E = I_1 w_1^2 + I_2 w_2^2 + I_3 w_3^2$$

defines 2 surfaces which are ellipsoids. Intersection of these 3 surfaces will give a line. This line determines the set of values w_1, w_2, w_3 can take. If we plot these surfaces considering w_1, w_2, w_3 as the xyz coordinates of this w-space.

Case 1: $w_1 \neq 0$; $w_2, w_3 \approx 0$

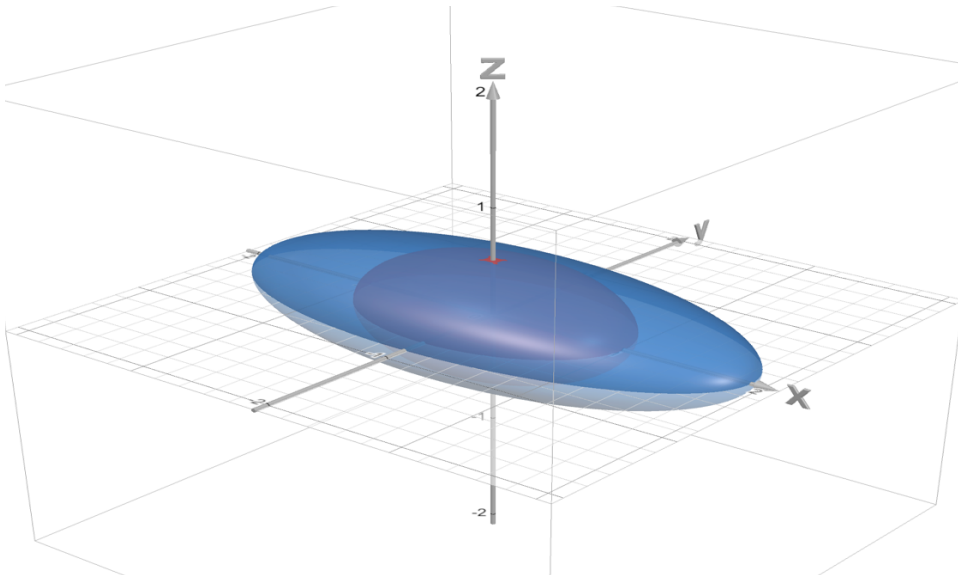
In this case ellipsoids intersect around a region where w_2, w_3 are very small (large axis). If we plot this using an online tool:



As one can see intersection occurs at a local region. w_2, w_3 are bounded to stay in this region and they cannot diverge any further. So one can conclude in this case rigid body is stable.

Case 2: $w_3 \neq 0$; $w_2, w_1 \approx 0$

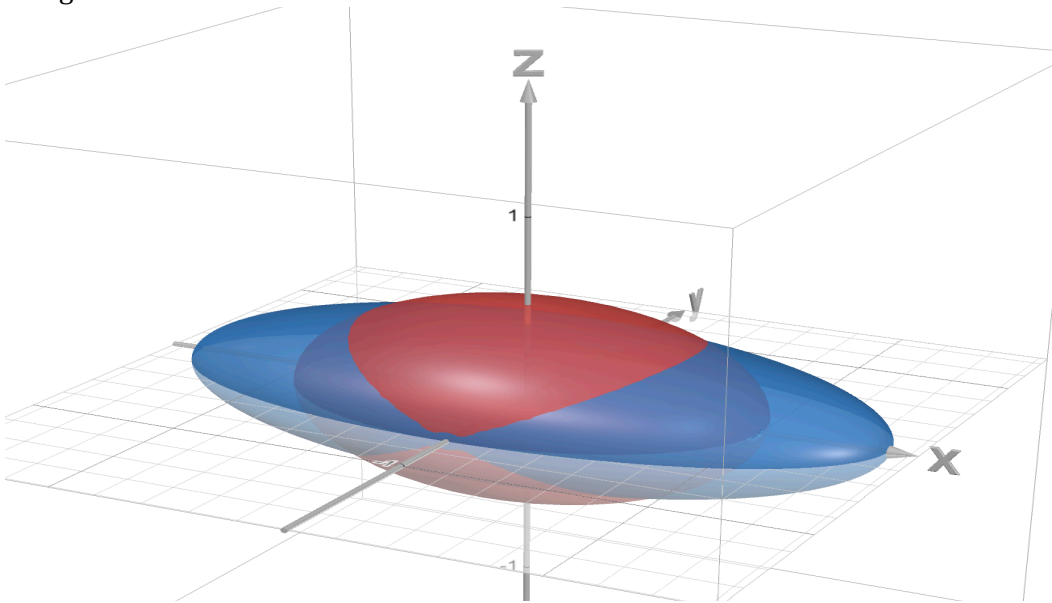
In this case ellipsoids intersect around a region where w_2, w_3 are very small (small axis). If we plot this using an online tool:



Again intersection occurs at a local region. w_2, w_1 are bounded to stay in this region and they cannot diverge any further. So one can conclude in this case too, rigid body is stable.

Case 3: $w_2 \neq 0$; $w_3, w_1 \approx 0$

In this case ellipsoids intersect around a region where w_1, w_3 are very small (middle axis). If we plot this using an online tool:



Contrary to the last 2 cases w_1, w_3 are not bounded to stay in a local region. Instead they can now take a wide range of values. They will not stay small but rather grow unboundedly. Then, one can conclude rigid body is **not** stable around the axis that has middle moment of inertia.